

G. Lejeune Dirichlet's Werke. II.

XV.

On a Property of Quadratic Forms of Positive Determinant.

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English translation

On a Property of Quadratic Forms of Positive Determinant.

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Since the new theorem which forms the subject of this lecture is closely connected with the theory of the equation

$$t^2 - Du^2 = 1, \tag{1}$$

some remarks must first be made about this equation. Here the given positive integer D is not to be equal to a square, and we shall have to consider only those solutions which are expressed by positive integers t, u .

Let T, U be the smallest values satisfying (1). Then, as is well known, all the solutions of the kind just mentioned are obtained from the formula

$$(T + U\sqrt{D})^n = t_n + u_n\sqrt{D}$$

by assigning to the integer n all positive values. Among these solutions there are infinitely many in which u_n is divisible by an arbitrary positive number S , and the exponents n for which this circumstance occurs are the successive multiples of the smallest such exponent N . Indeed, put

$$D' = DS^2$$

and form the new equation

$$t'^2 - D'u'^2 = 1.$$

It is then clear that every solution of this latter equation gives, when one puts

$$t = t', \quad u = Su',$$

a solution of (1) in which u is divisible by S , and conversely. This proves the assertion and also shows that N is determined by the equation

$$(T + U\sqrt{D})^N = T' + U'\sqrt{D'},$$

where T', U' denote the smallest values of t', u' .

One can specify the exponent N without knowing T', U' , as soon as, for each of the distinct prime factors p contained in S , one knows the smallest value ν for which u_ν is divisible by p , and also the exponent δ of the highest power of p by which u_ν is divisible. It is not hard to see that, if e denotes an arbitrary number divisible by p^ε , and by no higher power of p , under the assumption just made $p^{\delta+\varepsilon}$ will be the highest power of p occurring in $u_{\nu e}$. Therefore the necessary and sufficient condition for $u_{\nu e}$ to be divisible by p^α is that e be divisible by $p^{\alpha-\delta}$, where of course the difference $\alpha - \delta$, if it becomes negative, is to be replaced by zero.

Now distinguish by indices the various prime factors p contained in S , together with their corresponding values ν, δ , and write

$$S = p_1^{\alpha_1} p_2^{\alpha_2} \dots .$$

Then, by what has just been said, N is the least common multiple of the numbers

$$\nu_1 p_1^{\alpha_1 - \delta_1}, \quad \nu_2 p_2^{\alpha_2 - \delta_2}, \quad \dots .$$

It follows at once that, if without admitting new primes into S one lets all the exponents $\alpha_1, \alpha_2, \dots$ grow beyond every bound, the quotient S/N will soon reach a fixed value no longer depending on $\alpha_1, \alpha_2, \dots$.

The property just proved has an interesting consequence for the theory of quadratic forms of positive determinant. Add to the preceding assumptions the further assumption that D contains no square factor. Let h denote the number of classes into which the forms belonging to determinant D are divided, and let h' have the analogous meaning for the determinant $D' = DS^2$. Then, as was proved in an earlier paper, one has the equation

$$h' = h \frac{\log(T + U\sqrt{D})}{\log(T' + U'\sqrt{D})} SR,$$

where, as regards the factor R , it is to be observed that it depends on the primes $p_1, p_2, \dots$, but not on the exponents $\alpha_1, \alpha_2, \dots$. Since this equation may also be put in the form

$$h' = h \frac{S}{N} R,$$

it follows that from every determinant D infinitely many other determinants DS^2 can be derived, all of which have the same number of classes. By a suitable choice of D and of the primes $p_1, p_2, \dots$, one can arrange that this unvarying number of classes agrees with the number of genera. This proves the correctness of Gauss's conjecture that the series of positive determinants which possess only one class in each corresponding genus does not break off, whereas the negative determinants having the same property seem to exist only in finite number [*Disquisitiones arithmeticae*, art. 304].